JavaScript Complete Guide

Version 1.0 - PrepAI Knowledge Base

1. Introduction

JavaScript is a high-level scripting language used to build interactive web applications. It runs in browsers and on servers using Node.js.

2. Basics

Variables (var, let, const), data types, operators, type conversion.

3. Control Flow

if, else, switch, for, while, do-while, break and continue.

4. Functions

Function declarations, expressions, arrow functions, callbacks, scope and closures.

5. Objects & Arrays

Object literals, arrays, destructuring, spread operator, methods.

6. DOM Manipulation

Selecting elements, events, event listeners, modifying HTML and CSS dynamically.

7. ES6 Features

Template literals, let/const, arrow functions, promises, async/await, modules.

8. Asynchronous JavaScript

Callbacks, Promises, async/await, Fetch API and API requests.

9. Error Handling

try, catch, finally and custom errors.

10. Interview Questions

- Difference between var, let and const?
- What is hoisting?
- Explain closures.
- What is the event loop?
- Difference between == and ===?
- What are promises?
- What is async/await?
- What is the DOM?
- What is event bubbling?
- Explain callback functions.

11. Mini Project

Build a To-Do List, Weather App or Calculator using HTML, CSS and JavaScript.

12. Best Practices

Use const by default, modular code, meaningful variable names, avoid global variables, handle errors properly.